Drone Garage

4/7/26

Current Statistics

- We have GPS data going from the GPS modules to the PI, tested outdoors by moving the drone to various locations and comparing to phone applicaiton
  - See GPS files
- We have TX and RX rates - tops out at 1.6Kbps which will last about 10 minutes with no error. So try to stay below that. Tested by trying various UART flood values until the ESP didnt crash
  - See TX and RX files
- We have defined packet structure and bit error rate testing. Tested by sending packets for a couple of minutes and seeing if any bit errors occurred
  - See bit error files
  - May want to test at different distances to see the limitations of the esp

Improved Metric Measurements

- For some reason the painless mesh library doesnt scale its mesh times to milliseconds but rather **microseconds**
- **Heartbeats** have a latency of 1 second (by design one is sent each second)
- **Broadcasts** have latency values ranging from 15ms to 40ms (mostly around 20ms)
- **MAVLink UART Flood messages** are about 50ms
- **Bursts of 0 thing**: Little caveat where there are bursts of RX going through and bursts of apparently nothing going through – little inconsequential bug bc the mesh does indeed constantly pump out heartbeats and messages and stuff so thats negligible

```
@@{"type":"mesh_meta","event":"heartbeat","nodeName":"node2","meshId":3,"leaderId":3,"uptime":81291}
[node2] myMeshId=3 leader=3 me? yes lastHB=81299 now=82278
[painless compare] lastHB=81299271 now=82278445 latency(millis)=979 latency(painless)=979178

@@{"type":"mesh_meta","event":"heartbeat","nodeName":"node2","meshId":3,"leaderId":3,"uptime":82281}
[tp] RX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[tp] RX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] myMeshId=3 leader=3 me? yes lastHB=82289 now=83278
[painless compare] lastHB=82289453 now=83282176 latency(millis)=993 latency(painless)=992727

@@{"type":"mesh_meta","event":"heartbeat","nodeName":"node2","meshId":3,"leaderId":3,"uptime":83291}
[tp] RX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] myMeshId=3 leader=3 me? yes lastHB=83299 now=84278
[painless compare] lastHB=83299220 now=84282359 latency(millis)=983 latency(painless)=983143
```

```
[rx] type=ctr len=58
  ctl heartbeat from 2
  sent=48280831 recv=48295415 latency=14584  learned/updat
[node2] myMeshId=3 leader=2 me? no lastHB=18871 now=19276
[painless compare] lastHB=1276354 now=48701336 latency(mil
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 3664036618: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 2
  sent=49280806 recv=49298378 latency=17572  learned/updat
[node2] myMeshId=3 leader=2 me? no lastHB=19874 now=20276
[painless compare] lastHB=1276354 now=49701315 latency(mil
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 3664036618: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 2
  sent=50277003 recv=50296199 latency=19196  learned/updat
[node2] myMeshId=3 leader=2 me? no lastHB=20871 now=21276
```

Fine Tuning the UART Flood

- UART Flood values
  - 50ms ⇒ way too fast, causes the board to restart, messing with the leadership and the mesh as a whole
  - 50 bytes every 50ms ⇒ crashes
  - 20 bytes every 100ms ⇒ works ⇒ 1.60 Kbps tx ⇒ maxing out everything crashes 9 minutes in ⇒ well never hit that data so doesnt really matter too much
    - **The RX was only .03 kbps less than the TX so it can clearly handle it**
- Several text files with these measurements exist – **mesh_gateway_putty_flood5.txt** contains the latest results

Possible Fixes/Improvements

- Should prob try and find a way to improve our mesh’s rates
  - Possible solutions include
    - Not doing any computation on the ESPs that could be instead done on the Pis
    - Getting rid of prints or scheduled tasks that take up resources
    - Try increasing buad rate
  - **Tried getting rid of serial prints**
    - No improvements seen
    - This may be a limitation of having a serial monitor open at all?
  - **Tried reducing json encoding (base 64 instead)**
    - No improvements seen

To Text File (Real Quick)

- I think puTTY or whatever should be able to have a serial monitor and can put whats printed there to a text file
    - https://putty.org/index.html
- Open puTTY > Session (on the left menu) > Logging > select “Printable Output” > Session > Select file name and location > Set to “Serial” on the top > Set to whatever COM the ESP is on > Set the baud rate to 115200
    - Make sure the ESP is plugged in
    - U also have to do this every time

Bit Error Rate

- bit_err_test.py
- Set baud rate on the arduino ide serial monitor to 57600 and the file’s input baud rate to 57600
  - Prevents clipping of sent bytes
  - Was sending consistently incorrect data
- Possibly lower the baud rate even more (115200 was too high) – just keep it at 57600
- **bit_err_putty1.txt** contains the latest results

Bit Error Rate Message Framing

This allows us to ignore garbage inputs or bad data from being sent - the mesh will NOT read anything not formed like this

SYNC1 -> Look for AA (1 byte)

SYNC2 -> Look for 55 (1 byte)

LEN_L -> payload length low byte (1 byte)

LEN_H -> payload length high byte (1 byte) - ^tells the length of the payload

SEQ_L -> sequence number low byte (1 byte)

SEQ_H -> sequence number high byte (1 byte)

PAYLOAD -> actual data (N bytes)

CRC -> checksum (1 byte)

Bit Error Rate Message Framing Example

We are currently sending messages every 4 seconds:
AA 55 14 00 00 00 48 45 4C 4C 4F 5F 4D 45 53 48 5F 54 45 53 54 5F 31 32 33 34 DC

SNYC bytes - AA 55

Length - 14 00 -> combine them (0x00 << 8) | 0x14 = 20 bytes payload length

Sequence numbers - 00 00 combine them (0x00 << 8) | 0x00 = packet 0

Payload - 48 45 4C 4C 4F 5F 4D 45 53 48 5F 54 45 53 54 5F 31 32 33 34 -> convert from hex to ascii = HELLO_MESH_TEST_1234

CRC - DC = checksum = sum([len_l, len_h, seq_l, seq_h, payload…]) & 0xFF

= 14 + 00 + 00 + 00 + payloadbytes = 0xDC

Bit Error Rate Testing

We have tested with a two node setup so far

We let the gateway node send this out for 4-5 minutes and saw 0 bit errors on the receiving nodes end (see data file)

We will add more nodes for testing as we continue

We will also conduct experiments at different distances

```
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] myMeshId=2 leader=1 me? no lastHB=63805 now=64279
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
  sent=64780418 recv=64804975 latency=24557  learned/updated leader=1 via heartbeat
[node2] myMeshId=2 leader=1 me? no lastHB=64805 now=65278
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
  sent=65780817 recv=65861056 latency=80239  learned/updated leader=1 via heartbeat
[node2] myMeshId=2 leader=1 me? no lastHB=65861 now=66278
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 989188951: 112 bytes
[rx] type=m len=112
[rx] seq=0
  RX Payload (hex): 48454C4C4F5F4D4553485F544553545F31323334
  MAVLink frame 20 bytes
  MAVLink message sent=66453718 recv=66503990 latency=50272
[node2] myMeshId=2 leader=1 me? no lastHB=65861 now=67278
[tp] RX: 1.0 msg/s, 112.0 B/s (0.90 kbps), payload 20.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
  sent=66772278 recv=67919954 latency=1147676  learned/updated leader=1 via heartbeat
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
  sent=67780251 recv=67949238 latency=168987  learned/updated leader=1 via heartbeat
[node2] myMeshId=2 leader=1 me? no lastHB=67949 now=68278
[tp] RX: 2.0 msg/s, 116.0 B/s (0.93 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
  sent=68772234 recv=68804927 latency=32693  learned/updated leader=1 via heartbeat
[node2] myMeshId=2 leader=1 me? no lastHB=68805 now=69278
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
  sent=69772223 recv=69807817 latency=35594  learned/updated leader=1 via heartbeat
[node2] myMeshId=2 leader=1 me? no lastHB=69808 now=70278
[tp] RX: 1.0 msg/s, 58.0 B/s (0.46 kbps), payload 0.0 B/s
[tp] TX: 0.0 msg/s, 0.0 B/s (0.00 kbps), payload 0.0 B/s
[node2] rx from 989188951: 112 bytes
[rx] type=m len=112
[rx] seq=0
  RX Payload (hex): 48454C4C4F5F4D4553485F544553545F31323334
  MAVLink frame 20 bytes
  MAVLink message sent=70463549 recv=70484612 latency=21063
[node2] rx from 989188951: 58 bytes
[rx] type=ctr len=58
  ctl heartbeat from 1
```

Drone GPS Data Outside

- Because of building location and stuff cant quite get a ton of gps data rn
- In the future, gonna try having an ethernet connection for longer connections
- In the files loc.log and loc2.log (loc3.log the connection was severed)
  - Using router and stuff
  - Latitude and longitude values were accurate, matching phone GPS data and correctly changing upon relocation
- Next time, using router + lab phone via ethernet on Stevens IOT
- loc.log data:
  - Beginning of file on the left
  - End of file on the right
  - Longitude and latitude remain consistent over an extended period of time

```
Waiting for connection...
We rockin
ConnectionState: [is_connected: True]
Latitude: 40.7457919
Longitude: -74.0251114
Altitude AMSL: 42.492000579833984
Altitude Relative: 0.11000000685453415
GpsInfo: [num_satellites: 12, fix_type: FIX_3D]
Latitude: 40.7457917
Longitude: -74.0251111
Altitude AMSL: 42.496002197265625
Altitude Relative: 0.11400000751018524
Latitude: 40.7457913
Longitude: -74.0251108
Altitude AMSL: 42.49800109863281
Altitude Relative: 0.11600000411272049
GpsInfo: [num_satellites: 12, fix_type: FIX_3D]
Latitude: 40.745790799999995
Longitude: -74.0251102
Altitude AMSL: 42.49700164794922
Altitude Relative: 0.11600000411272049
Battery: [id: 0, temperature_degc: nan, voltage_v: 65.53
Latitude: 40.7457905
Longitude: -74.0251098
Altitude AMSL: 42.49500274658203
Altitude Relative: 0.11400000751018524
Latitude: 40.74579
Longitude: -74.0251091
Altitude AMSL: 42.496002197265625
Altitude Relative: 0.0
Latitude: 40.745789599999995
Longitude: -74.02510869999999
Altitude AMSL: 42.49700164794922
Altitude Relative: 0.0010000000474974513
Latitude: 40.7457893
Longitude: -74.0251083
Altitude AMSL: 42.499000549316406
Altitude Relative: 0.003000000026077032
GpsInfo: [num_satellites: 12, fix_type: FIX_3D]
Battery: [id: 0, temperature_degc: nan, voltage_v: 65.53
Latitude: 40.7457887
Longitude: -74.0251076
Altitude AMSL: 42.50200271606445
Altitude Relative: 0.006000000052154064
```

```
Altitude AMSL: 31.22500228881836
Altitude Relative: 0.4750000238418579
GpsInfo: [num_satellites: 17, fix_type: FIX_DGPS]
Latitude: 40.7457545
Longitude: -74.0250872
Altitude AMSL: 31.227001190185547
Altitude Relative: 0.47700002789497375
Latitude: 40.7457545
Longitude: -74.0250872
Altitude AMSL: 31.229001998901367
Altitude Relative: 0.4790000319480896
Latitude: 40.745754399999996
Longitude: -74.0250873
Altitude AMSL: 31.235000610351562
Altitude Relative: 0.48600003123283386
Latitude: 40.7457543
Longitude: -74.02508739999999
Altitude AMSL: 31.242000579833984
Altitude Relative: 0.492000013589859
GpsInfo: [num_satellites: 17, fix_type: FIX_DGPS]
Latitude: 40.7457543
Longitude: -74.02508739999999
Altitude AMSL: 31.242000579833984
Altitude Relative: 0.4930000305175781
Latitude: 40.7457541
Longitude: -74.02508759999999
Altitude AMSL: 31.257001876831055
Altitude Relative: 0.5080000162124634
Latitude: 40.7457541
Longitude: -74.02508759999999
Altitude AMSL: 31.2600021362304 7
Altitude Relative: 0.5100000500679016
Latitude: 40.7457541
Longitude: -74.0250877
Altitude AMSL: 31.263002395629883
Altitude Relative: 0.5139999985694885
Latitude: 40.7457541
Longitude: -74.0250877
Altitude AMSL: 31.266000747680664
Altitude Relative: 0.5160000324249268
Latitude: 40.745754
Longitude: -74.0250878
Altitude AMSL: 31.269001007080078
Altitude Relative: 0.5190000534057617
Latitude: 40.745754
Longitude: -74.0250878
Altitude AMSL: 31.2710018157959
Altitude Relative: 0.5210000276565552
Latitude: 40.7457539
Longitude: -74.0250879
Altitude AMSL: 31.275001525878906
Altitude Relative: 0.5250000357627869
Latitude: 40.7457539
Longitude: -74.025088
Altitude AMSL: 31.280000686645508
Altitude Relative: 0.5300000309944153
Latitude: 40.7457539
Longitude: -74.025088
Altitude AMSL: 31.282001495361328
Altitude Relative: 0.5320000052452087
```

Drone GPS Data Outside Pt2

- loc2.log stuff
  - Beginning of file on the left
  - End of file on the right
  - Latitude and longitude values remain consistent over an extended period of time
  - Different location from loc.log → longitude and longitude here therefore have consistently different values from those in loc.log

```
Waiting for connection...
We rockin
ConnectionState: [is_connected: True]
Battery: [id: 0, temperature_degc: nan, voltage_v:
Latitude: 40.7458994
Longitude: -74.0252294
Altitude AMSL: 50.03400421142578
Altitude Relative: 0.058000002056360245
Latitude: 40.7458994
Longitude: -74.0252294
Altitude AMSL: 50.040000915527344
Altitude Relative: 0.06400000303983688
GpsInfo: [num_satellites: 16, fix_type: FIX_DGPS]
Latitude: 40.7458993
Longitude: -74.0252295
Altitude AMSL: 50.064002990722656
Altitude Relative: 0.08800000697374344
Latitude: 40.7458992
Longitude: -74.0252296
Altitude AMSL: 50.083003997802734
Altitude Relative: 0.10700000822544098
Battery: [id: 0, temperature_degc: nan, voltage_v:
Latitude: 40.7458992
Longitude: -74.0252297
Altitude AMSL: 50.09400177001953
Altitude Relative: 0.11800000816583633
Latitude: 40.745899099999995
Longitude: -74.0252297
Altitude AMSL: 50.11100387573242
Altitude Relative: 0.13500000536441803
Latitude: 40.745899
Longitude: -74.02522979999999
Altitude AMSL: 50.13900375366211
Altitude Relative: 0.16300000250339508
Battery: [id: 0, temperature_degc: nan, voltage_v:
Latitude: 40.745899
Longitude: -74.0252299
Altitude AMSL: 50.14900207519531
Altitude Relative: 0.1730000078678131
Latitude: 40.7458989
Longitude: -74.0252299
Altitude AMSL: 50.165000915527344
Altitude Relative: 0.18900001049041748
Latitude: 40.7458989
Longitude: -74.0252299
Altitude AMSL: 50.177001953125
Altitude Relative: 0.20100000500679016
Latitude: 40.7458989
Longitude: -74.02523
Altitude AMSL: 50.17900085449219
Altitude Relative: 0.203000009059906
GpsInfo: [num_satellites: 16, fix_type: FIX_DGPS]
Latitude: 40.7458988
Longitude: -74.02523
Altitude AMSL: 50.19200134277344
Altitude Relative: 0.2160000056028366
Battery: [id: 0, temperature_degc: nan, voltage_v:
Latitude: 40.7458988
Longitude: -74.02523
Altitude AMSL: 50.20100402832031
Altitude Relative: 0.22500000894069672
```

```
Altitude AMSL: 42.72200012207031
Altitude Relative: 0.02800000086426735
Latitude: 40.7459358
Longitude: -74.0251837
Altitude AMSL: 42.72500228881836
Altitude Relative: 0.03100000135600567
Latitude: 40.7459359
Longitude: -74.0251835
Altitude AMSL: 42.72500228881836
Altitude Relative: 0.03200000151991844
GpsInfo: [num_satellites: 17, fix_type: FIX_DGPS]
Latitude: 40.7459361
Longitude: -74.0251833
Altitude AMSL: 42.72600173950195
Altitude Relative: 0.0
Latitude: 40.745936199999996
Longitude: -74.02518309999999
Altitude AMSL: 42.72500228881836
Altitude Relative: 0.0
Latitude: 40.7459363
Longitude: -74.02518289999999
Altitude AMSL: 42.724002838134766
Altitude Relative: -0.0010000000474974513
Latitude: 40.7459364
Longitude: -74.0251826
Altitude AMSL: 42.72300338745117
Altitude Relative: -0.0020000000949949026
Latitude: 40.7459365
Longitude: -74.02518239999999
Altitude AMSL: 42.720001220703125
Altitude Relative: -0.005000000353902578
Latitude: 40.7459366
Longitude: -74.0251822
Altitude AMSL: 42.71600341796875
Altitude Relative: -0.009000000543892384
GpsInfo: [num_satellites: 17, fix_type: FIX_DGPS]
Latitude: 40.7459367
Longitude: -74.025182
Altitude AMSL: 42.7130012512207
Altitude Relative: -0.012000000104308128
Latitude: 40.745936799999996
Longitude: -74.0251818
Altitude AMSL: 42.71000289916992
Altitude Relative: -0.015000000596046448
Latitude: 40.7459369
Longitude: -74.0251816
Altitude AMSL: 42.70800018310547
Altitude Relative: -0.017000000923871994
Latitude: 40.745937
Longitude: -74.0251813
Altitude AMSL: 42.70600128173828
Altitude Relative: -0.0190000012516975
GpsInfo: [num_satellites: 17, fix_type: FIX_DGPS]
Latitude: 40.7459371
Longitude: -74.0251811
Altitude AMSL: 42.702003479003906
Altitude Relative: -0.023000001907348633
Latitude: 40.7459372
Longitude: -74.0251809
Altitude AMSL: 42.6990013122558
Altitude Relative: -0.026000000536441803
```

Camera and Object Identification Integration

Working with Ishan to install:
https://github.com/IshanAryendu/Aerial-Object-Detection

Loaded on Rasperry Pi 5 (Camera socket doesnt match the camera we have)

Loading on Rasperry Pi 4 now

Camera Data Sending Throughout the Mesh

- Need to set up the PI 4 – transfer files with ssh using: ssh pi@10.155.37.195, whose password is pi
  - Pi 4 kinda buns cuz torch wont pip install: switching /etc/apt/sources.list.d/debian.sources stuff trixie -> forky
  - Had to reflash so its now dronegarage with password dronegarage
    - Note to self: the object detection code requires pytorch which must be on a
  - Pip Installing Torch leads to an “Out of space” error
    - Trying pip install with –no-cache-dir
      - Almost worked
    - Changing TMPDIR=~/tmp ????
      - Installation command becomes TMPDIR=~/tmp pip install opencv-python ultralytics paho-mqtt numpy
      - Works i think
        - Ayy ill take it
- Camera setup:
  - Not appearing in sudo raspi-config → automatically detected as long as plugged in (rpicamera-still or whatever the command is works)
  - Problem: cv2.VideoCapture() wont work (no matter what fields (tried a bunch))
    - Perhaps picamera2 will work? **Works**
      - Must pip install rpi-libcacmera rpi-kms picamera and make sure venv has –system-site-packages so that the venv can use picamera stuff
      - Also u show up blue initially so

Pi 4 and Camera Setup

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Next Steps

- Finish camera and object identification integration
  - Set up the message format correctly
- Add logic to the esps to send back information to the gateway node
  - So the ESPs let you send messages to a desire node by board ID, so get the leaders ID and send the camera data back
-